A Multimodal Fusion Fatigue Driving Detection Method Based on Heart Rate and PERCLOS

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***Abstract*— Existing visual-based fatigue detection methods usually monitor drivers' fatigue by capturing their facial features, including eyelid movements, yawn frequency and head pose. However, these approaches typically do not take drivers' biological signals into consideration. An accurate model for fatigue detection requires combining both facial behavior and biological data. This paper proposes a novel non-intrusive method for driver multimodal fusion fatigue detection by extracting eyelid features and heart rate signals from the RGB video. The multimodal feature fusion method could significantly increase the accuracy of fatigue detection. Specifically, we established two fatigue detection models based on heart rate and the PERCLOS value respectively with one-dimensional Convolutional Neural Network (1D CNN), where the PERCLOS refers to the percentage of eyelid closure over the pupil. Finally, the outputs of the two models are weighted to achieve the multimodal fusion fatigue detection. Simulation results show that our method yield better performance than traditional methods.**

***Index Terms*— Multimodal feature fusion, fatigue driving detection, PERCLOS, heart rate.**

I. INTRODUCTION

LONG driving hours or insufficient rest can cause diver fatigue, which will greatly increase the probability of danger. Studies have shown that approximately 25%-30% of traffic accidents are from driver's insufficient rest [1]. Fatigue driving detection allows us to remind the fatigue driver to take a rest, thereby avoiding dangerous situations, and is crucial for traffic safety. Fatigue driving detection methods can be broadly classified into five types: subjective reports, physical signals, biological signals, the vehicular features, and hybrid features [2]. Due to the randomness of subjective reports and the difficulty of obtaining vehicle data, we introduce fatigue features from the following three aspects.

In visual-based fatigue detection method, physical features are extracted from facial expression data acquired by a camera. Facial expressions, including mouth and eyes, can provide valuable information about the driver's status [3]. Yawning is a sign of sleepiness or boredom, and Bhandari [4] used the yawning videos to detect fatigue. In addition, when a driver feels fatigued, blink frequency may become faster, and the maximum closure duration of the eyes may become longer [5]. In previous, Singh [6] determined the driver is fatigued if the time of the driver's eyes closed is more than a certain threshold. Although it is feasible sometimes, not all drivers will close their eyes for a long time when they are tired. In recent years, efforts have been made to seek for alternative driving fatigue measurements. Among them, PERCLOS (the percentage of eyelid closure over the pupil) is one of the most effective visual-based approaches [7]. And the smaller the PERCLOS value is, the faster the eye-opening speed is. Mandal [8] used spectral regression to map the image of the eye area with the degree of the eye closure, and then calculated PERCLOS through RGB video. Wang [9] introduced Gabor filter and vectorized local integral projection to process pictures, and then built a GP-CNN focusing on eye image processing to obtain blink frequency and PERCLOS. Dasgupta [10] proposed a three-stage fatigue detection system based on mobile phones. However, this application needs to run on a mobile device with high processing performance, and lacks precision due to only single eye feature captured. Some studies [11]–[14] used both eye and mouth features for fatigue detection. Savaş [11] classified fatigue results into three categories. Li *et al.* [12] established a personalized classifier based on the driver. Liu [13] combined the static and dynamic features of the face to classify the fatigue. In [11] and [14], CNN was used for eyes and mouth feature extraction and fatigue classification. However, the more facial features are not the higher accuracy. Huang [15] performed multi-granularity feature extraction on faces and the features at a reasonable proportion of participation in decision-making. It illustrated the importance of the proportion of participating features and the importance of facial feature fusion. At the same time, the size of the fatigue detection model is also considered in the application. Reddy [16] compressed the model and the input features. The system can be deployed on low-cost embedded devices.

Fatigue is a nerve activity accompanied by weak biological signals. Biological signals including electrocardiogram (ECG),

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electroencephalogram (EEG), electromyography (EMG) and electrooculogram (EoG), are considered as more reliable features because they are generally not controlled by consciousness [17]. Fu [18] tried to apply an ECG for fatigue detection. They used the cross curve of surface EMG and ECG peak factors as features for fatigue detection. Oviyaa [19] developed a drowsiness detection system that has EEG-sensor within a helmet to realize automatic alarming and motor deceleration for bikers. Fan [20] introduced forehead EEG to detect fatigue. Extracted features including energy, entropy, rhythmic energy ratio and frontal asymmetry ratio were fed into the machine algorithms. Besides, Wu [21] mapped the EEG signals and obtained LTM-BPM (Long Term Memory-Brain Power Map) which can accurately express brain fatigue information. And then input it into CNN-LSTM model for brain fatigue detection. In addition, some studies [21]–[23] showed that the heart rate variability (HRV) feature can also reflect the behavior of subjects. There are two HRV acquisition methods: ECG and photoplethysmography (PPG) [17]. The HRV features were extracted from the ECG and then were fed into different deep learning classifiers, which shows the superiority of deep learning in predicting fatigue status with HRV features [22]. Chang [23] analyzed HRV in the time domain, frequency domain, trend, approximate entropy, and sample entropy, and finally input them into the neural network for fatigue classification. With the development of contactless heart rate measurement, Nowara [24] obtained reliable heart rate estimation from narrow-band near-infrared video recordings. The remote PPG (rPPG) has become a topic of interest [25].

Note that all the above methods rely on a single modal data, which limits the adaptation to various scenes and the reliability of model prediction. And each parameter has its pros and cons and therefore it is encouraged to use their combination [26]. Some methods have combined multimodal data for fatigue detection, which involves the extraction of multimodal features and the fusion models. For example, Sun [27] used a multiclass support vector machine to perform a two-level fusion of contextual features. In [28]–[30], they used two different modal features and indicated the hybrid features fatigue system can be used to decrease accidents. PERCLOS feature and EEG feature were integrated in [28] and facial feature and Haar-like feature for face and vehicle were used in [29]. Bakker [30] combined the physical data of the eyes and head pose with vehicle data to detect fatigue, achieving high accuracy in the test set. Besides, the fatigue detection module with the fusion of three different modal features was used in the automated driving system [31]. However, some of them ignored the fact that a person's fatigue state is continuous, a better scheme needs to extract feature variations over a period of time. And non-intrusive is of significance as the sensor used to get other features will disturb the driver. Although the above methods have achieved good results, they can be further improved due to the following reasons:

1) Conventional visual-based driving fatigue detection methods do not take drivers' biological signals into consideration, combining drivers' biological signals and facial behavior facilitates fatigue detection in a non-intrusive way.
2) Fatigue status is a continuous time series data. Some existing methods focus on processing the features of a certain moment, ignoring the changes of fatigue features over time.
3) Most existing methods of obtaining fatigue features require the driver to stay in a special environment or wear a sensor, which is costly to deploy and not friendly to the drivers.

To address those limitations, we develop a multimodal fusion fatigue detection method based on heart rate and PERCLOS. Inspired by [32], we also use facial features and heart rate to detect fatigue status. However, our methods replace the depth camera with a normal camera for the sake of deployment cost. Instead of using eye openness level and mouth openness level, PERCLOS is one of the most effective fatigue detection features in facial expression. Furthermore, our work is different from [32], we adopt a different deep learning model with a multimodal fusion strategy. Specifically, we first study the calculation of PERCLOS and use the existing method to obtain rPPG signal based on RGB videos. Then, we use two 1D CNNs to respectively train the two fatigue detection models based on PERCLOS and heart rate signal. Finally, the final fatigue result is obtained by fusing the two models. The main contributions of this paper can be summarized as follows.

1) We propose a non-intrusive driving fatigue detection approach, which takes the drivers' biological signals into account and avoids the discomfort caused by wearable devices.
2) To improve the performance of driver fatigue detection, two fatigue detection tasks, i.e., PERCLOS-based fatigue detection and heart rate-based fatigue detection, are unified under a fusion method.
3) A novel low-cost deployment approach, based on RGB video to obtain iris using K-means clustering, is deployed to approximate PERCLOS. To better classify the fatigue status, we set a sliding window with 30 frames to extract fatigue features over a period of time and 1D CNN is used to detect fatigue instead of setting a threshold for PERCLOS.

This paper is organized as follows. Section II introduces an overview of the fatigue detection framework. Section III illustrates the methods for extracting PERCLOS and heart rate. Section IV demonstrates the structure of the fatigue detection model. The experimental environment and results are presented in Section V. The last section provides a conclusion and discusses further work.

II. Overview

The overall framework for multimodal fusion fatigue detection is shown in Fig. 1. It mainly includes face detection and tracking, extraction of two signals, model construction, and model fusion. Firstly, we initialize the area of the face detected in the first frame and the face feature points. Then, we initialize the sliding windows for counting PERCLOS and rPPG signals respectively. Subsequently, face detection detects the local area

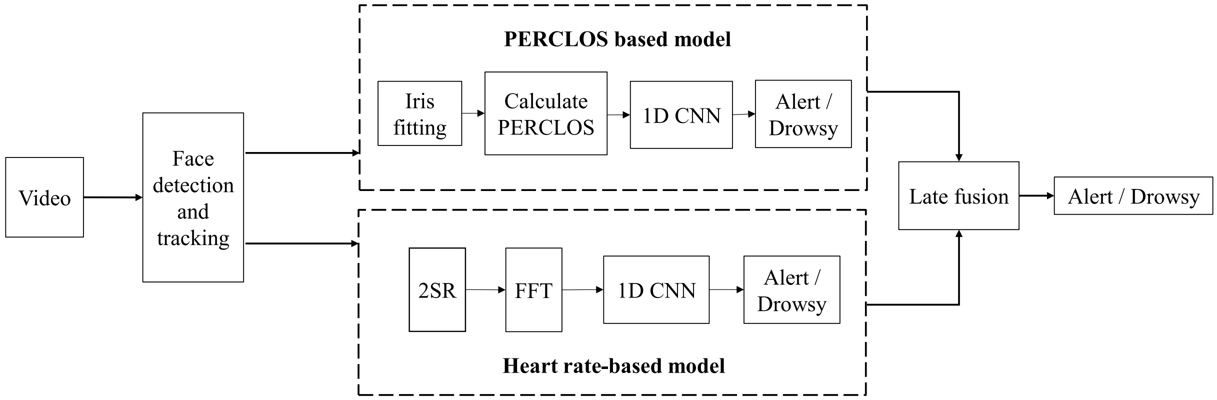

Fig. 1. The overall framework for multimodal fusion fatigue detection. After face detection, the video frames are segmented to get iris and then calculate PERCLOS. At the same time, the rPPG signals are obtained from the face video by 2SR algorithm. The two features are respectively sent to train the two models. The final result is obtained by the late fusion.

predicted by KLT optical flow tracking [33] instead of the next complete frame, which can save a lot of detection time.

(1) PERCLOS based model. We propose a segmentation-based algorithm to detect the iris instead of the traditional pupil detection method. It is possible to calculate the PERCLOS value of the driver through RGB video without using an infrared camera and wearing special glasses. We identify the shape of the iris in the eye image to determine whether the eye is closed and calculate the PERCLOS value through the sliding window. Then we use a 1D CNN to extract the fatigue features over a period of time and classify the fatigue.

(2) Heart rate-based model. According to the Spatial Subspace Rotation (2SR) [34] algorithm, we can obtain the rPPG signal by calculating the rotation of the skin pixels of the face in the adjacent frames in the RGB space, which has less noise compared with the other methods. We use a sliding window to intercept the rPPG signal and then employ the Discrete Fourier Transform to get the frequency of the signal which can be converted to heart rate. We also use a 1D CNN to detect the fatigue based on heart rate change over a period of time.

(3) Fusion of two models. We separately train PERCLOS based model and heart rate-based model, then we obtain the final fatigue status by weighting the result of these two models.

III. Fatigue Features Extraction

This section describes how to extract PERCLOS and heart rate. The used data are facial videos captured by an RGB camera. Part A mainly introduces how to approximate the PERCLOS value through the iris recognition with an unsupervised learning method. Part B introduces how to get rPPG and describes the steps of heart rate conversion.

A. PERCLOS Extraction

An important indicator for studying the feature of eyelid movement is PERCLOS, which refers to the ratio of the time that the area of the eyelid covering the pupil exceeds a certain threshold in the statistical interval [30]. Since we use the RGB camera instead of the infrared camera, this paper will approximately convert the area of the eyelid covering the pupil into the area of the eyelid covering the iris. At the same time, it is considered that the area of the eyelid covering 50% of the iris is equivalent to the area of the eyelid covering 70% to 80% of the pupil. We propose an iris detection algorithm based on unsupervised learning that checks whether the eye is closed according to the percentage of eyelid coverage. And PERCLOS value is the ratio of the closed duration to the duration of the statistical interval.

The flowchart for extracting the boundary of iris is shown in Fig. 2. First, we use the face detection algorithm to obtain the eye area. Then the grayscale image of the eye is binarized with OTSU algorithm. In addition, all the pixels in RGB images of this region are segmented using K-means, where the value of hyperparameter K is 4. Since K-means is an unsupervised learning method and each pixel is assigned to a cluster without label, we cannot locate the specific pixel cluster of the iris only by K-means. We perform pixelwise clustering in the binary image. The specific positioning method is to calculate the IoU (Intersection Over Union) value of each pixel cluster and the black cluster of the binary image. Because the iris is black in the binary image, the pixel cluster is an iris when the IoU value of a certain pixel cluster is relatively large. Furthermore, we perform Canny edge detection on the image area after the iris is located. Since the shape of the iris is elliptical under different angles, we use detected curve segments for ellipse fitting. Then the position and shape of the fitted ellipse is seen as that of the iris.

Since the frame rate of the image collected by the camera is constant, there is a linear relationship between the number of image frames and the length of the video. A comprehensive four steps approach for PERCLOS calculation is deployed and is presented in Fig. 3 and elaborated on in Table I. The sliding window has 30 frames and the step size is 1 frame. The overlapping frames for two consecutive windows are 29 frames. Due to the last frame is newly added and the other frames in the queue have already calculated the eyelid closure, we only use the last frame to obtain the eyelid closure each time. After calculating the eyelid closure of the last frame, the PERCLOS can be obtained by calculating the ratio of the closed frames to the total frames in the queue.

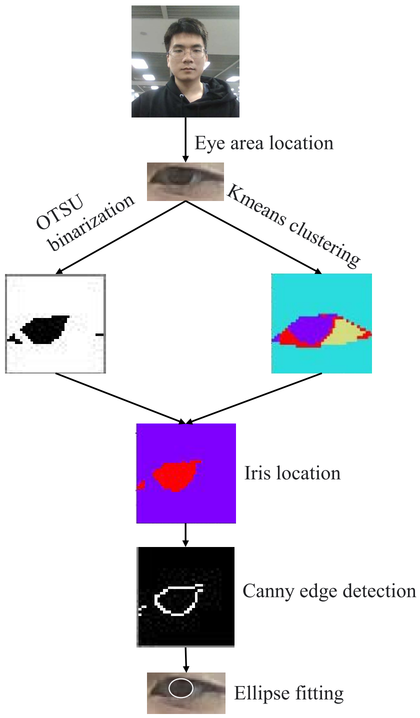

Fig. 2. Flowchart for extracting the boundary of iris.

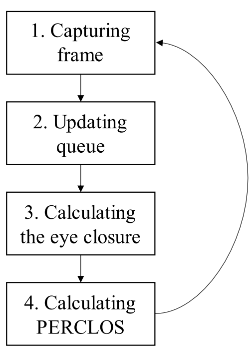

Fig. 3. The approach for PERCLOS calculation.

TABLE I
DESCRIPTION OF THE STEPS USED TO CALCULATE PERCLOS

Steps	Operations
1. Capturing frame	Capturing a new face image frame from the video.
2. Updating queue	If the sliding window queue is full, remove the first frame of the queue. Then, add the new frame.
3. Calculating the eyelid closure	If the eyelid covers the iris more than 50%, increase the number of the closed frame by one.
4. Calculating PERCLOS	Getting PERCLOS by counting the ratio of the closed frames to the total frames in the sliding window.

B. Heart Rate Extraction

To achieve the non-intrusive heart rate measurement, we propose to use the rPPG method which captures the visible light or infrared light of the face with a camera and then analyzes the frequency to get the heart rate. There are two steps to get heart rate. First, getting the skin area of the face. Skin videos are obtained from human faces because there are generally no covers. Second, extracting the rPPG signal. The heart rate can be obtained by calculating the frequency of the rPPG.

We use the KLT to track the Harris feature points of the driver's face. The basic idea of the KLT is as follows: when a rigid body moves under the same lighting condition, the pixel intensity of two adjacent frames in the camera is constrained. The pixel points can be tracked by calculating the displacement of the same pixel point in two adjacent frames. The KLT method has two assumptions: (1) The tracked object is on a rigid body. In other words, there is no relative displacement around the tracked pixel. (2) The pixel intensity does not change, which means the pixel value of the same pixel point in the different frames does not change. Combining these two assumptions, we can get that the gradient value between a pixel point, and the surrounding pixel does not change in the two adjacent frames.

Here are two methods to acquire rPPG signal: CHROM [35] and 2SR.

CHROM use a self-adaptive method to estimate the mapping vector and reduce the sensitivity of priori information. First, a standard skin tissue vector and corresponding mapping matrix is defined as follows:

$$M^{-1} * u_s = 1, \tag{1}$$

where u_s represents a 3×1 average skin tissue vector under white light. M represents the diagonal mapping matrix, which is used to map $\tilde{C_n}(t)$ as:

$$M * \tilde{C_n}(t) = M * 1 * i(t) + M * N * u_s * I_0 * s(t) + M * N * u_p * I_0 * p(t). \tag{2}$$

The normalized color of skin is assumed to be equal to the color of standard skin tissue under white light. Therefore, the spectral reflection vector $N * u_s * I_0$ is approximately mapped as

$$M * N * u_s * I_0 \approx k * 1, \tag{3}$$

where k is a scaling factor. The CHROM will map $M * \tilde{C_n}(t)$ to a plane which is orthogonal to the unit vector.

$$S(t) = P_c * M * \tilde{C_n}(t) \approx P_c * M * 1 * i(t) + P_c * M * N * u_p * I_0 * p(t), \tag{4}$$

where $P_c * M * N * u_s * I_0 \approx k * P_c * 1 = 0$ and P_c is an 2×3 initial projection matrix. Each line of P_c is the projection axis, which defines a plane in the RGB space. In the CHROM, $P_c * M$ is a vector with fixed value.

Then, the corresponding rPPG signal is calculated as follows:

$$p(t) = S_1(t) - \alpha * S_2(t), \tag{5}$$

where $\alpha = \frac{\sigma(S_1)}{\sigma(S_2)}$.

The 2SR algorithm takes the changes of RGB values of skin in different frames into consideration. The study found that the rPPG signal can be obtained by calculating the rotation of the skin pixel in the RGB space. 2SR first calculates the correlation coefficient matrix of skin pixel on a frame. Then, the coordinate system corresponding to the correlation coefficient matrix is obtained by decomposing the matrix. Finally, the rPPG signal can be obtained by calculating the rotation of each coordinate system on two frames. The skin pixels are converted into RGB space. It can be expressed by the following formula:

$$C = \frac{P^T * P}{n}, \tag{6}$$

where n represents the number of skin pixels. The dimension of P is $n \times 3$. The dimension of C is 3×3, which represents the correlation coefficient matrix of these pixels in the RGB space. We can express C through a coordinate system and make $C = x * i + y * j + z * k$. Then, we can get the rotation between the coordinate systems. In this way, the question is how to obtain a coordinate system to represent the matrix C.

The idea of 2SR is to obtain an orthogonal matrix by decomposing the matrix C. The column vectors of an orthogonal matrix are the three axis vectors of the corresponding coordinate system. It can be represented as follows:

$$C * U = \lambda * U, \tag{7}$$

where λ is the eigenvalue diagonal matrix of C and U is the eigenvector matrix of C. Then, the matrix C can be expressed as follows:

$$C = \lambda * U * U^T. \tag{8}$$

As U is an orthogonal matrix, $U^{-1} = U^T$. According to the above formula, the linear combination of the coordinate axis vector of C can be represented as

$$C = \lambda_1 * u_1 * u_1^T + \lambda_2 * u_2 * u_2^T + \lambda_3 * u_3 * u_3^T, \tag{9}$$

where λ_i represents the value of the ith row and ith column of the matrix λ. Now we can get (u_1, u_2, u_3) in the RGB space for the skin pixels of each frame.

Then, the rPPG signal can be obtained by calculating the rotation between the coordinate systems of adjacent frames. Assume that the coordinate system of the current frame is U_t and the next frame is U_{t+1}, the rotation between U_t and U_{t+1} can be expressed as follows:

$$R_{t\to t+1} = U_t^{\mathrm{T}} * U_{t+1} = \left(u_1^t, u_2^t, u_3^t\right)^{\mathrm{T}} * \left(u_1^{t+1}, u_2^{t+1}, u_3^{t+1}\right). \tag{10}$$

We can obtain the rotation matrix of the skin pixels of adjacent frames. We mark θ_{ij} as the angle of rotation between u_i^t and u_j^{t+1}. According to the angle relationship between the inner product of the two vectors, we can get $\cos\left(\theta_{ij}\right)$.Since both u_i^t and u_j^{t+1} are unit vectors, the rotation angle can be obtained by substituting into each element of the rotation matrix.

$$R_{t\to t+1} = \begin{bmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) & \cos(\theta_{13}) \\ -\cos(\theta_{12}) & \cos(\theta_{22}) & \cos(\theta_{23}) \\ -\cos(\theta_{13}) & -\cos(\theta_{23}) & \cos(\theta_{33}) \end{bmatrix} \tag{11}$$

The obtained rotation angle θ is the rPPG signal.

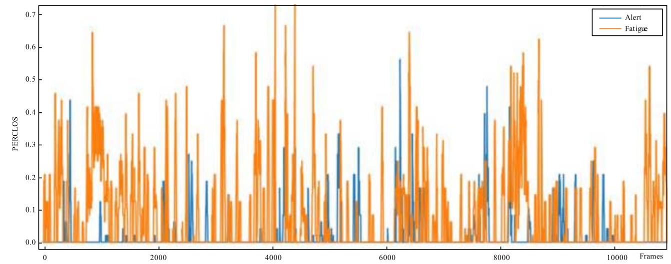

Fig. 4. Drivers' PERCLOS under alert and fatigue states.

The acquired rPPG signal is $P(t)$. It is a one-dimensional signal that changes over time, reflecting the change of blood volume. We use the short-time fast Fourier transform to extract the heart rate. The steps are as follows:

Step 1: We perform frequency noise filtering on the obtained rPPG signal. Due to the heart rate is generally maintained in the range of 40 to 170, a 0-25Hz filter is used to eliminate the low frequency part. Then, we use a 1D CNN to smooth the signal and filter out the tiny glitch noise signals between the adjacent peaks.

Step 2: We use a sliding window to intercept the rPPG data and then use the fast Fourier transform to calculate the frequency of the rPPG in the sliding window.

Step 3: We select the frequency with the largest amplitude as the frequency of the rPPG signal. Then, we choose the frequency within 60 seconds to get the heart rate. Heart rate conversion formula is represented as follows:

$$HR = f * 60\,(bpm). \tag{12}$$

IV. The Fatigue Detection Model Establishment

We use the method in Section III to obtain PERCLOS and heart rate. In this Section, Part A and B establishes a fatigue detection model based on PERCLOS and a fatigue detection model based on heart rate, respectively. Part C introduce the boosting method to fuse the two models.

A. Fatigue Detection Model Based on PERCLOS

As shown in Fig. 4, there are many large peaks in the corresponding PERCLOS signal when the driver is in the fatigue state. We can count the frequency of wave crests to identify fatigue. However, this threshold is not easy to set when fatigue is uncertain. And a higher PERCLOS value may be caused by the driver looking down.

In addition to considering peaks in PERCLOS, it is also necessary to consider the signals around the peak. We use a sliding window to intercept the PERCLOS signal and detect whether the driver is fatigue through the changes of time series data. The shallow convolutional neural network can learn local features that are independent of position. And the PERCLOS signal fits this situation exactly. The peaks of the PERCLOS signal appear randomly, but its fluctuation is fixed. Since the PERCLOS and heart rate feature are represented by 1D signal and 1D CNN has the time and performance advantages in dealing with 1D signals, we used 1D CNN for fatigue detection.

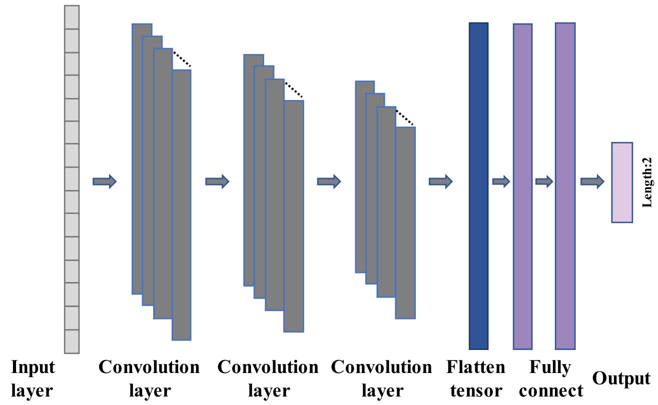

Fig. 5. The fatigue detection model based on 1D CNN.

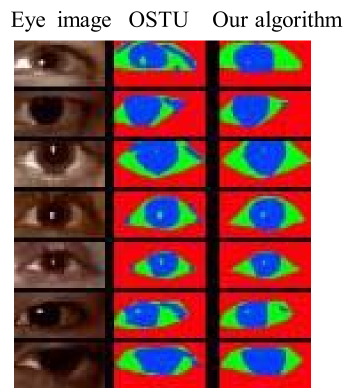

Fig. 6. Comparison results between the OTSU algorithm and the proposed binarization algorithm.

Our proposed 1D CNN architecture for fatigue detection is shown in Fig. 5. It consists of an input layer, three convolution layers, and two fully connected layer. The size of the input is 512×1. There are three convolution layers with filter length of 24 and each layer uses the ReLU activation function. The size of convolution kernels at convolutional layer 1 is 8×1, the one at layer 2 is 4×1, and the one at lay 3 is 4×1. At the end of the network, we add two fully connected layer (FCL) to perform the classification. The two FCL layers has 128 and 64 neurons respectively. To prevent over-fitting, we use SoftMax classifier which computes the probability for the two fatigue states. The value of the network is initialized by assigning random values.

B. Fatigue Detection Model Based on Heart Rate

At present, many studies have shown that fatigue can be well detected based on heart rate variability (HRV). HRV refers to the change in heart rate over a period. The current HRV based fatigue detection methods are to obtain ECG signals by attaching electrodes to the subjects' skin and then converting the signals into HRV. We cannot obtain the HRV directly from heart rate. Since the goal is to learn how the heart rate signal changes over time, the heart rate of sliding window over time also helps us achieve fatigue driving detection. Therefore, we designed a method to extract the change of heart rate at adjacent moments for fatigue detection and built a model based on 1D CNN.

We also use the network shown in Fig. 5. The parameters are different. The size of the input is 1024×1. There are three convolution layers with filter length of 32 and each layer uses the ReLU activation function. The size of convolution kernels at convolutional layer 1 is 16×1, the one at layer 2 is 8×1, and the one at lay 3 is 4×1. At the end of the network, we add two fully connected layer (FCL) to perform the classification. The two FCL layers has 256 and 128 neurons respectively. To prevent over-fitting, we use SoftMax classifier which computes the probability for the two fatigue states.

C. Multimodal Fusion Algorithm Based on Boosting Method

Due to the fatigue detection method is composed of two independent models. It means that either PERCLOS or heart rate signals can independently detect whether the driver is fatigued. In the statistical learning method, there is a model called Boosting method which is very suitable for integrating multiple weak precision sub-models into a strong precision model [36]. The Adaboost algorithm is a typical one in the boosting method. The basic idea of the boosting is to select some data from the training set to make one sub-model achieve sufficiently high accuracy, and then select some data to make the other sub-model achieve sufficiently high accuracy. Different training samples have different weights in different sub-models. The final output is the weighted outputs of these sub-models [37]. Therefore, the key to the boosting method lies in how to design the weights of the sub-model results. There are two main ideas in Adaboost. The first is to increase the weight of the misclassified samples in the next sub-model training process, so that different sub-models can cover as many samples as possible. The second is to make the sub-model with high accuracy more weighted and make the sub-model with low accuracy a smaller weight in the final result. Iterative training in this way until the error rate converges below the threshold.

We use the ideas of Adaboost. First, we train the fatigue detection model based on PERCLOS and the fatigue detection model based on heart rate respectively. Then, we calculate their classification error rate on the test set.

$$error_{rate} = \frac{1}{n}\sum\nolimits_{i=0}^{n} P(M(x_i) \neq y_i) \tag{13}$$

Based on this error rate, we calculate the weight of the two models in the final result. The formula of weight can be calculated as

$$\alpha_m = \ln\left(\frac{1 - error_{rate}}{error_{rate}}\right). \tag{14}$$

In this way, the accuracy of the fusion model is improved by reducing the weight of the model with a high error rate and increasing the weight of the model with a low error rate.

V. Experiments and Anylisis

To verify the performance of the proposed model, comparative experiments are carried out including eyelid feature extraction and verification, heart rate extraction, performance evaluation of multimodal fusion fatigue detection model and different models on the same dataset.

TABLE II
KSS TABLE [38]

Fatigue level	Description
1	Extremely Alert
2	Very Alert
3	Alert
4	Rather Alert
5	Neither Alert or Sleepy
6	Some signs of sleepiness
7	Sleepy, but no difficulty remaining awake
8	Sleepy, some effort to keep alert
9	Extremely Sleepy, fighting sleep

These experiments will use the fatigue video dataset (UTA-RLDD [38]) collected by the University of Texas under realistic scenes. The UTA-RLDD dataset is the largest fatigue video dataset under realistic scenarios until now. More importantly, the data is generated in an environment like the actual driving scenario. The dataset contains 60 healthy subjects which includes undergraduates, graduate students, staffs who volunteers to participate, and others who wants to get extra credits from courses at the school. And all the participants are over 18 years old, which fits the actual scene of fatigue driving very well. Among them, there are 51 men and 9 women. They are between 20 and 59 years old with a standard deviation of 6. The dataset has 180 RGB videos and each video is about 10 minutes. There are 21 videos where the subject wears glasses. The video was recorded from different angles in different real environments and backgrounds. Each video was recorded by participants themselves using their mobile phones or web cameras and the camera is placed in a position of about one meter from the subject. These specifications are to ensure that the angle of recorded video is consistent with the angle of video obtained by placing the RGB camera on the dashboard of the car. Each video has a subject status label. And it uses the KSS quantitative scale (Karolinska sleepiness scale, KSS) to divide into three fatigue classes: (1) Alert: One of the first three states in KSS table. (2) Low Vigilant: It is stated in level 6 and 7 of Table II. (3) Drowsy: The last two states in Table II.

Our method solves the binary classification problem of samples. We use videos representing alert and drowsy from the dataset to train and test the models. The dataset is split into train and test dataset with a ratio of 7:3. It means that 84 videos from 42 people are the training set and 36 videos from 18 people are the test set. After about 50 epochs, the models reach convergence. According to the fusion method mentioned in Part C of Section IV, the weights of the two models based on PERCLOS and based on heart rate are 0.5904 and 0.4096 respectively.

A. Eyelid Feature Extraction and Verification Experiment

To verify the effectiveness of the PERCLOS extraction from RGB video, we carried out a comparison experiment between the proposed eye region binarization algorithm and the OSTU binarization algorithm on iris extraction. The idea of extracting the iris is to use Dlib to extract the eye area, and then binarize the eye area. Performing edge detection on the binarized image and we use an ellipse to fit the iris. From the following figure, the binary image obtained by using the OSTU algorithm will connect the iris region and the eyelid region, while the combination of the K-means and IoU evaluation tags can well separate the iris from other areas.

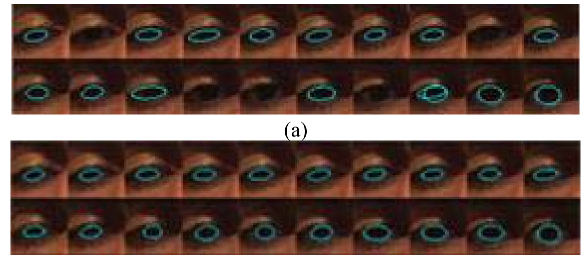

Fig. 7. (a) Iris fitting using the binary image obtained by OTSU. (b) Iris fitting using the binary image obtained by K-means and IoU evaluation.

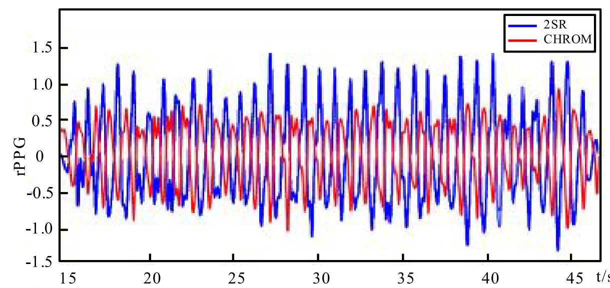

Fig. 8. rPPG signal extracted by 2SR algorithm and CHROM algorithm.

The reason for this case is that OTSU binarization only analyzes the pixel values of the grayscale image to calculate a threshold, so as to realize the binarization of the grayscale image. The K-means directly classifies each pixel according to the RGB value. In this way, it can comprehensively consider the three channels to cluster and segment the eye area. Since K is set to 4, 4 clusters are obtained. After segmentation through the RGB value, the label of iris cluster can be obtained by IoU evaluation. The iris region will be more accurate and not easily mixed with other regions. This makes it possible to fit the iris ellipse.

The next step is to perform Canny edge detection on the binary image of the iris area. From Fig. 7, we can see that our proposed method can fit the position and shape of the iris better. However, the ellipses after OTSU binarization often have problems such as insufficient accuracy, failure to detect the ellipse, and misdetection.

It can be verified that the combination of K-means and IoU evaluation is suitable for the iris detection problem. This is because the binarization method divides more finely and the iris area is not connected with other areas. In this way, the ellipse fitting will be more accurate according to the edge of the iris area.

B. Heart Rate Extraction Experiment

The size of the sliding window that extracts the rPPG signal is 30, which means that the rPPG signal is obtained by process-

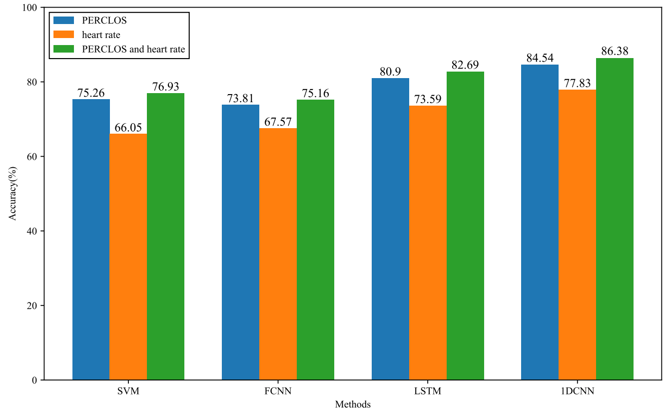

Fig. 9. Comparison of SVM, FCNN, LSTM and the 1D CNN under different feature combinations.

TABLE III

FOUR SITUATIONS OF FATIGUE DETECTION RESULTS

Label / Prediction	Alert	Drowsy
Alert	TP	FP
Drowsy	FN	TN

ing 30 frames in the adjacent moment each time. Fig. 9 is a fragment of the rPPG signal obtained from RGB video. The horizontal axis indicates that the time unit is seconds, and the vertical axis indicates the intensity of the rPPG signal. It can be seen from the figure that the 2SR algorithm can correctly extract the rPPG signal and has a clearer peak than that of CHROM.

According to the rPPG signal obtained by 2SR, we perform FFT on it to get the frequency. The heart rate corresponding to each moment can be calculated through the frequency.

C. Performance Evaluation Experiment of Multimodal Fusion Fatigue Detection Model

In this paper, PERCLOS signal and heart rate signal are used to train two fatigue detection models respectively. Then, according to the error rate of these two models, a multimodal fusion fatigue detection model is obtained. For the classification results of the above models, there are four situations as shown in Table III.

According to the above four situations, the accuracy and precision can be defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}, \tag{15}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{16}$$

Table IV shows that the multimodal fatigue detection model can detect fatigue more accurate than PERCLOS-based or heart rate-based fatigue detection model. It indicates that heart rate and PERCLOS do have complementary effects. It also verifies the effectiveness of the proposed fusion strategy.

In the experiment, the accuracy of fatigue detection based on heart rate is lower than that based on PERCLOS. The reason is that we obtained the rPPG signal through a visual-based method, which makes the obtained heart rate noisier.

TABLE IV

PERFORMANCE OF FATIGUE TESTING USING DIFFERENT FEATURES

Feature	Accuracy	Precision
Only PERCLOS	84.5%	85.1%
Only Heart rate	77.8%	78.4%
PERCLOS and heart rate	86.4%	87.1%

Therefore, it affects the fatigue detection accuracy. Generally, using a heart rate signal for fatigue detection is more accurate than PERCLOS. However, it occurs if the heart rate signal is obtained by electrodes and the heart rate noise is relatively small. In addition to the larger heart rate noise, another important reason is that there are strong individual differences in heart rate. And the test set completely contains 18 subjects. The data of these subjects did not participate in the training. The individual variability of PERCLOS is smaller than that of heart rate, so the accuracy of PERCLOS based detection modal is higher than that of heart rate based in the experiment.

In addition, we compare the accuracy of SVM, FCNN, LSTM and the 1D CNN when using different features for fatigue classification. Three models are trained with each method. Fig. 9 shows that the accuracy of any fusion model is higher than that of the model obtained by using only PERCLOS or heart rate. It means that our method combining PERCLOS and heart rate for fatigue detection can improve the accuracy of fatigue detection.

As for 1D CNN and LSTM, LSTM performance should be better than CNN in general when the input features are sequential data. Since our inputs are feature values rather than vectors, there are some different results. Then, some works [39] have presented an interpretation that the short-term temporal correlation is more significant than the long-term temporal correlation in eyelid movement. Our experimental results that the higher performance of CNN than LSTM also proves it. The accuracy of LSTM classifier is about 4% lower than CNN, which shows that 1D CNN is more suitable for fatigue detection based on PERCLOS and heart rate. Overall, our experiments also shows that when the extracted fatigue feature is numerical, CNN modeling is better than LSTM modeling.

D. Performance Evaluation Experiment of UTA-RLDD Dataset

To further verify the performance of the proposed method, we compared our method with HMLSTM [38], TFBI- LSTM [40], CNN [41] and LSTM [42] on the UTA-RLDD dataset. The performance of above methods on this dataset is illustrated in Table V. Our method is more accurate than the HMLSTM, TFBI-LSTM and LSTM methods in identifying fatigue. To achieve the goal of accuracy breakthroughs in fatigue detection for a single modal feature, it requires continuous research and optimization. And the fusion of multimodal features might yield better results under certain circumstances. According to the experiment in part C of section V, we can also see that the accuracy of using LSTM as a classifier is higher than the

TABLE V
THE COMPARISON OF DIFFERENT METHOD ON UTA-RLDD DATASET

Methods	Granularity extraction	Feature	Accuracy	Precision	Results
HMLSTM [38]	eyes	EAR, duration, Eye opening velocity, frequency	65.6%	/	✓physical features ×biological features ×Hybrid features ✓consider temporal dynamics
TFBI-LSTM [40]	eyes+mouth+nose	EAR, MAR, MVE, centre coordinate of puil	79.9%	80.6%	✓physical features ×biological features ×Hybrid features ×consider temporal dynamics
CNN [41]	face	Neural network extraction of features	91.8%	92.8%	✓physical features ×biological features ×Hybrid features ×consider temporal dynamics
LSTM [42]	heart rate	heart rate	75.83%	/	×physical features ✓biological features Hybrid features ✓consider temporal dynamics
Our Method	eyes+heart rate	PERCLOS, heart rate	**86.4%**	**87.1%**	✓physical features ✓biological features ✓Hybrid features ✓consider temporal dynamics

two methods in [38] and [40], which indicates the importance of fusion for different modal features. Although our accuracy is not higher than CNN [41], we consider the time sequence state of fatigue and biological features. In [41], it uses the random frame of the video as the image to classify fatigue which is not suitable for real-time video fatigue detection. And the features of a certain time point cannot completely reflect the fatigue state at this stage. What we need to obtain is the fatigue features over time, which can more accurately reflect fatigue status. Furthermore, when only heart rate is used as input, our result is higher than method [42] about 2%, which shows that convolutional neural network can also be used to process timing sequence data.

When there are many features as input, whether the input features are of same type, it needs to be combined and adjusted according to the importance of each feature to improve the detection accuracy. And we are inspired by the idea of boosting for late fusion. Finally, our method achieves two of the three general fatigue detection indicators: First, feasibility means low deployment cost. We can use the ordinary cameras or webcam for fatigue detection. Second, accuracy means that compared with other methods, the detection accuracy is relatively high. In terms of speed, since the extraction of the two features takes a certain time, simplified algorithm and reduced model parameters are considered.

As discussed in [43], some works are not exactly comparable when the experimentation methodology is different. Fatigue test criteria should be similar when compared with the baseline method, whether to classify a video or a frame image. Methods in [40], [41] all processes every frame of the video. Although the high accuracy can be achieved after development, the timing characteristics of fatigue are ignored. We consider that it would be necessary to evaluate its performance over the full duration of the videos. However, for fatigue video classification, there are not enough baseline methods. Compared with existing methods, our method still has certain advantages.

VI. CONCLUSION

This paper proposes a method for driver multimodal fusion fatigue detection by extracting the PERCLOS and heart rate signal from RGB video, which makes it possible to solve the problem of non-intrusively fusing the driver's eyelid movement feature and biological signals. For different features, we proposed different extraction methods and input them into a 1D CNN for training. Finally, the two models are fused based on the idea of boosting. Experimental results shows that the proposed method has good robustness and accuracy under certain conditions. Although it performs well in the existing dataset, the scenarios in the dataset are usually simpler than in the real world. Therefore, in the visual-based fatigue detection, the future of research and improvement is mainly to extract robust biological features obtained by learning-based method and combine them with other features. How to migrate the fatigue detection model to new subjects is also a problem to be solved.

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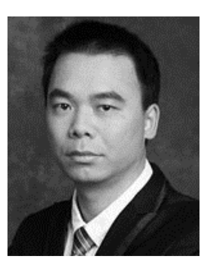

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